

Technical Document #833: Natural Language Processing - Comprehensive Overview

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Text summarization generates a concise summary of a longer text. Extractive summarization selects important sentences while abstractive summarization generates new text.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Sentiment analysis determines the emotional tone behind a body of text. It is widely used in social media monitoring and customer feedback analysis.

Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Sentiment analysis determines the emotional tone behind a body of text.
- Language models predict the probability of a sequence of words.
- Word embeddings represent words as dense vectors in continuous space.